D3COY 1.0 · DJS ISACA WITH C-DAC · PS-01

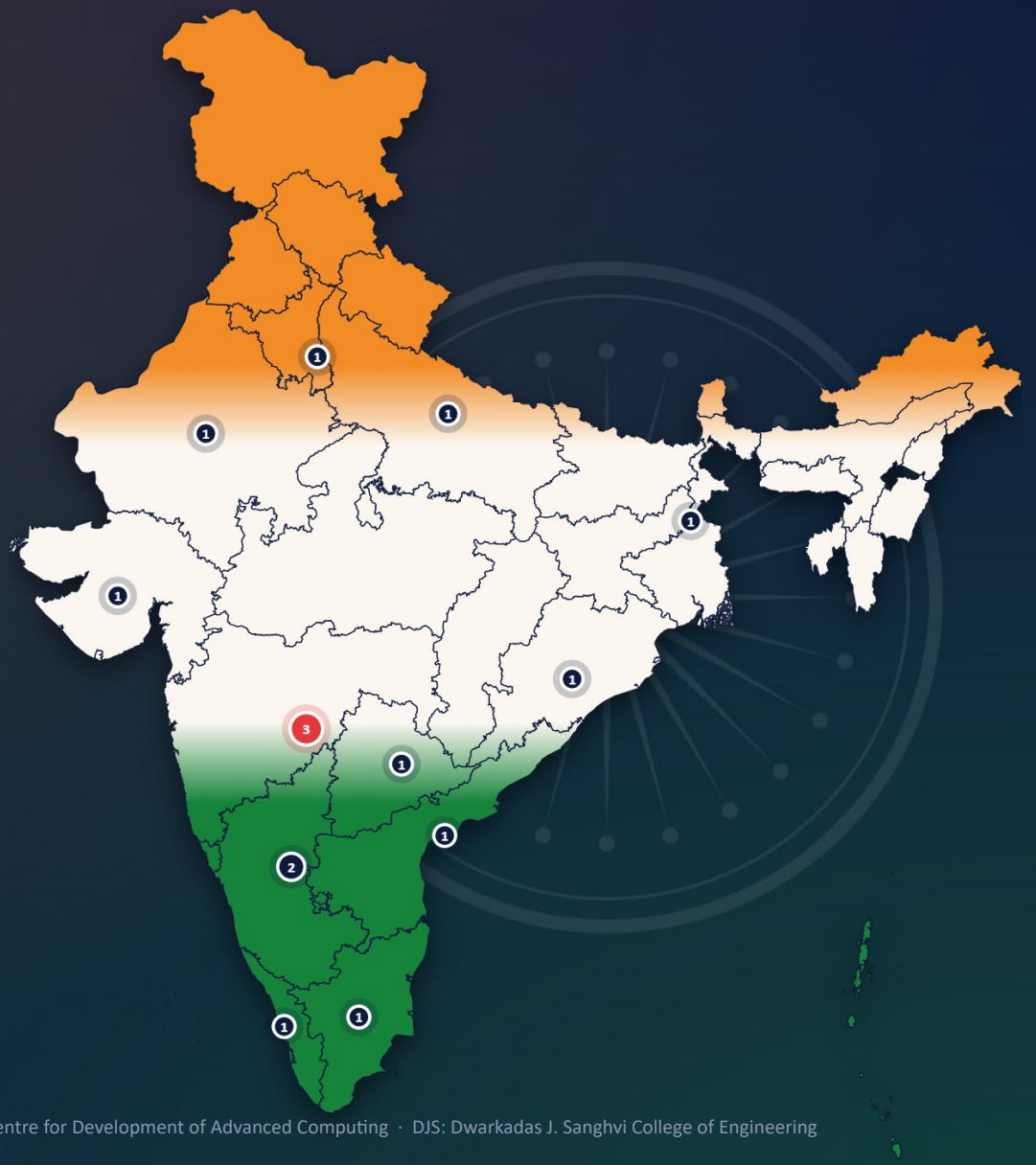
Odyssey

National cyber threat early warning, running entirely on one laptop.

It reads the public threat record and its own sensors, finds what threatens India's critical operators across six sectors and twelve states, and warns each in its own language, hours to weeks before the incident.

0 cloud calls **13** intelligence sources **22** Indian languages

Team Odyssey · 1 November 2026



On the map: the 15 operators we protect (simulated, labelled SIM), by state; the red one is the suspected compromise open right now.

Legend PS-01: Problem Statement 01 · SIM: simulated data, labelled as such · ISACA: Information Systems Audit and Control Association · C-DAC: Centre for Development of Advanced Computing · DJS: Dwarkadas J. Sanghvi College of Engineering

PROBLEM STATEMENT · PS-01

Security teams see the attack after it has happened.

Threats evolve faster than teams

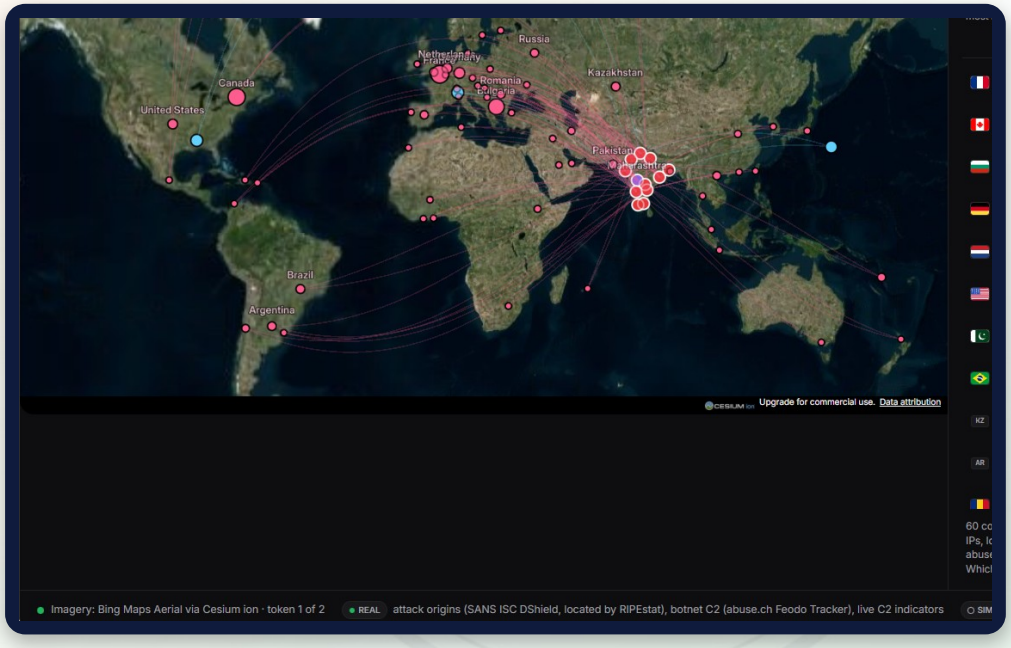
Newly discovered vulnerabilities, phishing campaigns, malware and attacks on digital infrastructure, every day.

Many sources, no single picture

Intelligence arrives from many feeds, so it is hard to spot what is emerging, correlate incidents and decide which systems come first.

Reaction, not anticipation

Traditional approaches react once suspicious activity has occurred. The brief asks for a proactive system that warns before threats become incidents.



Real attack origins this week: SANS ISC DShield's top 1,997 attacking IPs located by RIPEstat, reaching India's operators.

"A need for an intelligent, proactive system that can analyse available cyber threat intelligence, identify emerging attack patterns, and generate early warnings before threats develop into major security incidents." — the brief

Legend PS-01: Problem Statement 01 · DShield: SANS Internet Storm Center's distributed sensor network · SANS ISC: SANS Internet Storm Center

THE CHALLENGE

Turn raw threat information into three answers an analyst can act on.

1

What is happening?

Suspicious patterns, emerging campaigns and the systems they touch, correlated across every source.

2

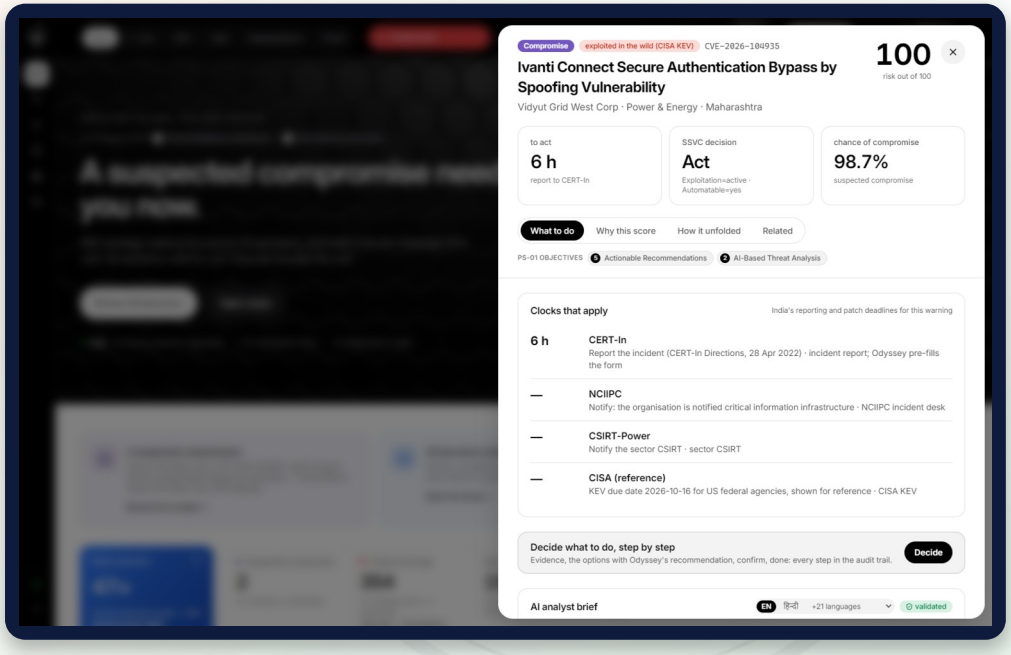
Which assets are at risk?

A new advisory, matched against the inventory: who runs the affected software, in which version, reachable from where.

3

What to do first?

A prioritised risk score, an early warning that explains itself, and the preventive measure to take now.



One warning answers all three: what it is and how urgent, which assets and versions, and what to do, with the clocks that apply.

Expected solution: a dashboard of incoming intelligence, an AI-generated threat summary, a prioritised risk score and an early warning that explains why. Odyssey ships all four, measured.

Legend PS-01: Problem Statement 01

WHAT WE BUILT

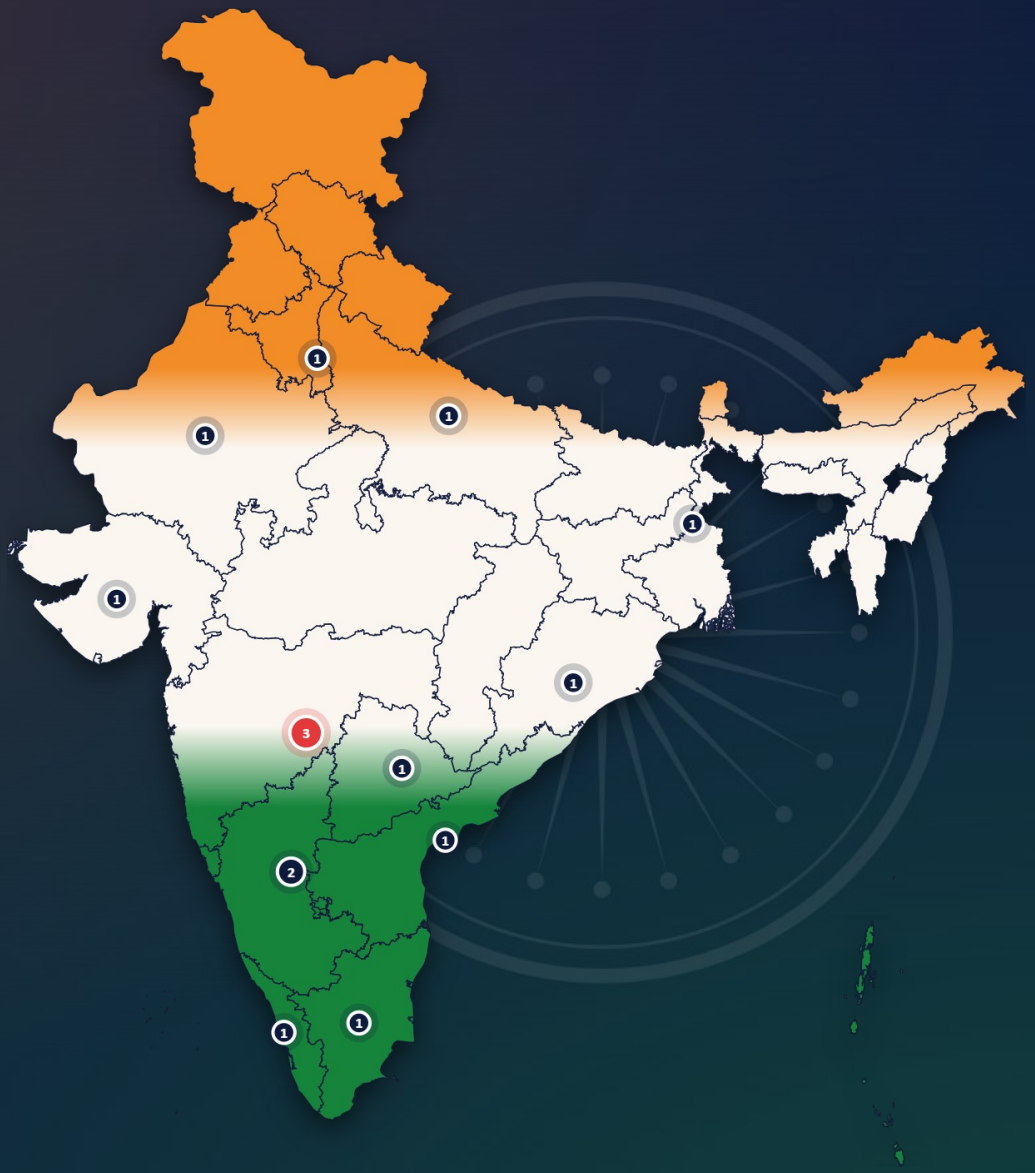
One early-warning loop for the nation's critical operators.

15 operators, SIM

6 NCIIPC sectors

13 intelligence sources

12 states



Power, banking, telecom, transport, government and strategic enterprises, each operator warned in its own language.

Legend NCIIPC: National Critical Information Infrastructure Protection Centre · SIM: simulated data, labelled as such

OUR SOLUTION

Odyssey closes the loop: six objectives, one pipeline, one audit trail.

1 Aggregate

13 sources, snapshot-backed and re-read live

2 Analyse

Security LLMs, SecureBERT, forecast, Bayes, detectors

3 Warn

NVD every 2 min, surges, day-0 flags, 22 languages

4 Prioritise

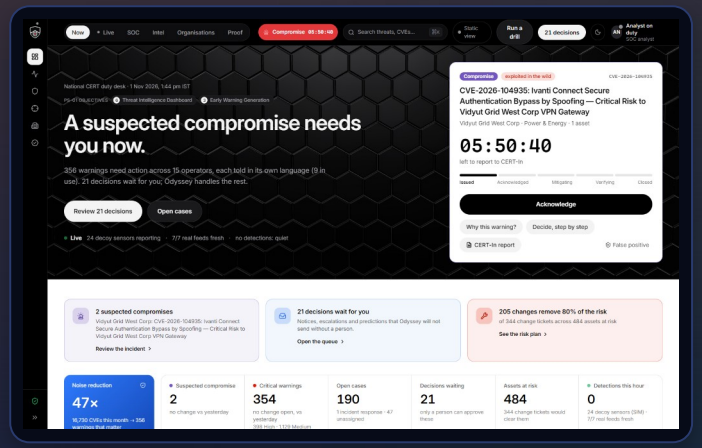
Risk = 100·R·K·(0.55L + 0.30S + 0.15V), SSVC, clocks

5 Act

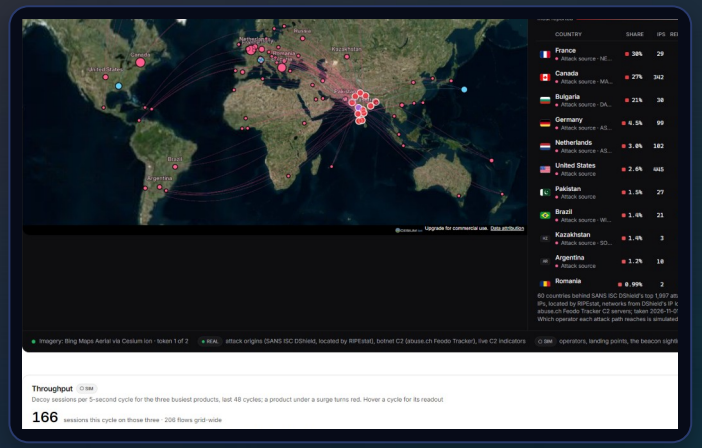
Fixed release, block, hunt, risk plan, decide step by step

6 See

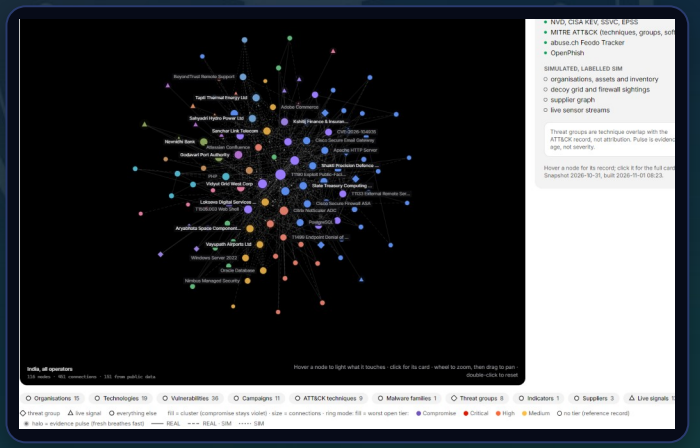
Duty desk, satellite map, SOC, graph, organisations, proof



Now: the duty desk



Live: the satellite threat map



Intel: the intelligence graph

Legend PS-01: Problem Statement 01 · NVD: National Vulnerability Database (NIST) · SSVC: Stakeholder-Specific Vulnerability Categorization · SOC: Security Operations Centre

THE SIX KEY OBJECTIVES

Each objective has a home in the product, and the screen says so.

1 **Threat Intelligence Aggregation**

In Odyssey: Intel › Overview, Live › Real feeds, Indicators in and out (STIX, TAXII)

2 **AI-Based Threat Analysis**

In Odyssey: The AI brief on every warning, advisory analysis, the intelligence graph, the day-0 forecast

3 **Early Warning Generation**

In Odyssey: The duty desk, the Live page (NVD intake, detectors, drills), SOC › Warnings

4 **Risk Prioritisation**

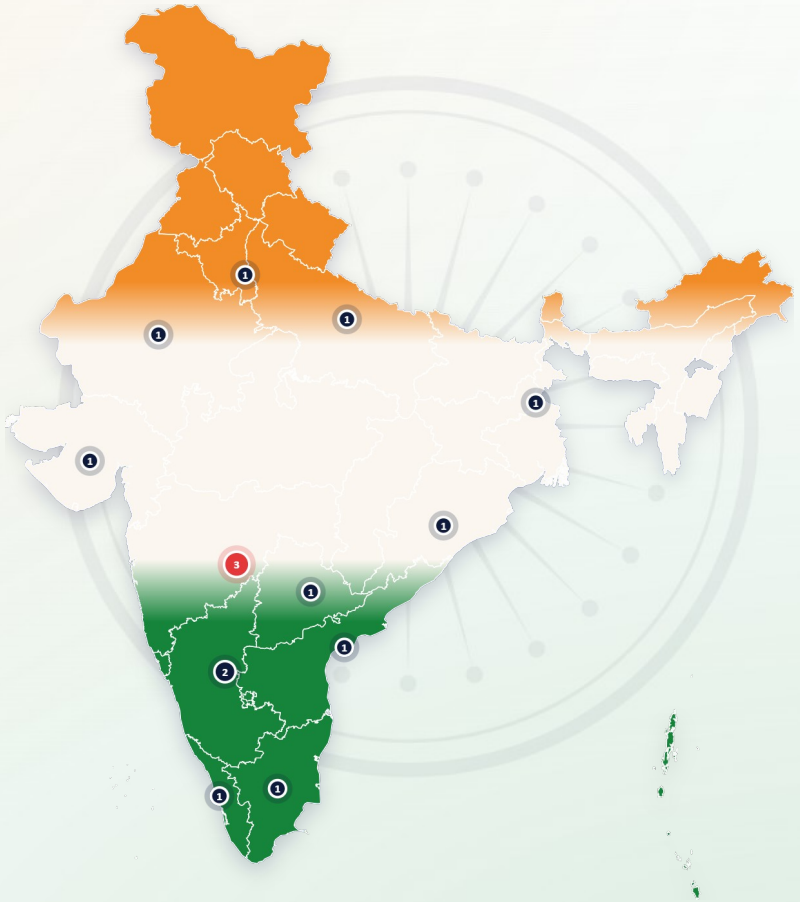
In Odyssey: Why this score on every warning, the ranked warning list, Intel › Risk plan, Organisations

5 **Actionable Recommendations**

In Odyssey: What to do on every warning, SOC cases with playbooks, change tickets, step-by-step decision support

6 **Threat Intelligence Dashboard**

In Odyssey: Now, SOC › Overview, Intel › Overview, Organisations, the satellite threat map



Operators by state (SIM); red: the suspected compromise

Every page, tab and warning sheet carries the pill of the objective it answers.

Legend PS-01: Problem Statement 01 · NVD: National Vulnerability Database (NIST) · STIX: Structured Threat Information Expression · TAXII: Trusted Automated Exchange of Intelligence Information · SOC: Security Operations Centre · SIM: simulated data, labelled as such

THE BRIEF'S EXAMPLE USE CASE, LIVE

A new advisory arrives. Within seconds, every exposed operator has a warning.

1. **Arrive**

   A real CISA ICS advisory for Siemens WinCC, or a CVE from NVD read every two minutes.

2. **Correlate**

   Version ranges parsed and checked against installed versions: 13 WinCC assets at 7 operators.

3. **Score**

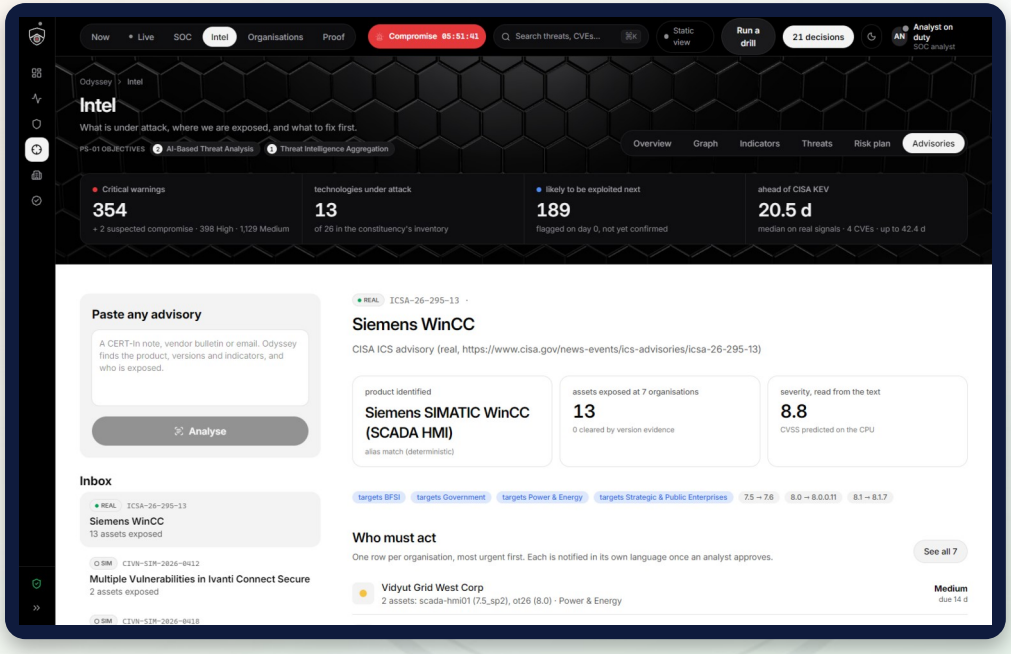
   Risk out of 100, an SSVC tier, the CERT-In deadline and a day-0 exploitation forecast.

4. **Warn**

   A validated AI brief, the named fixed release, indicators to block, the hunt steps.

5. **Decide**

   Notices go out in each operator's language once an analyst approves; every step is in the audit trail.

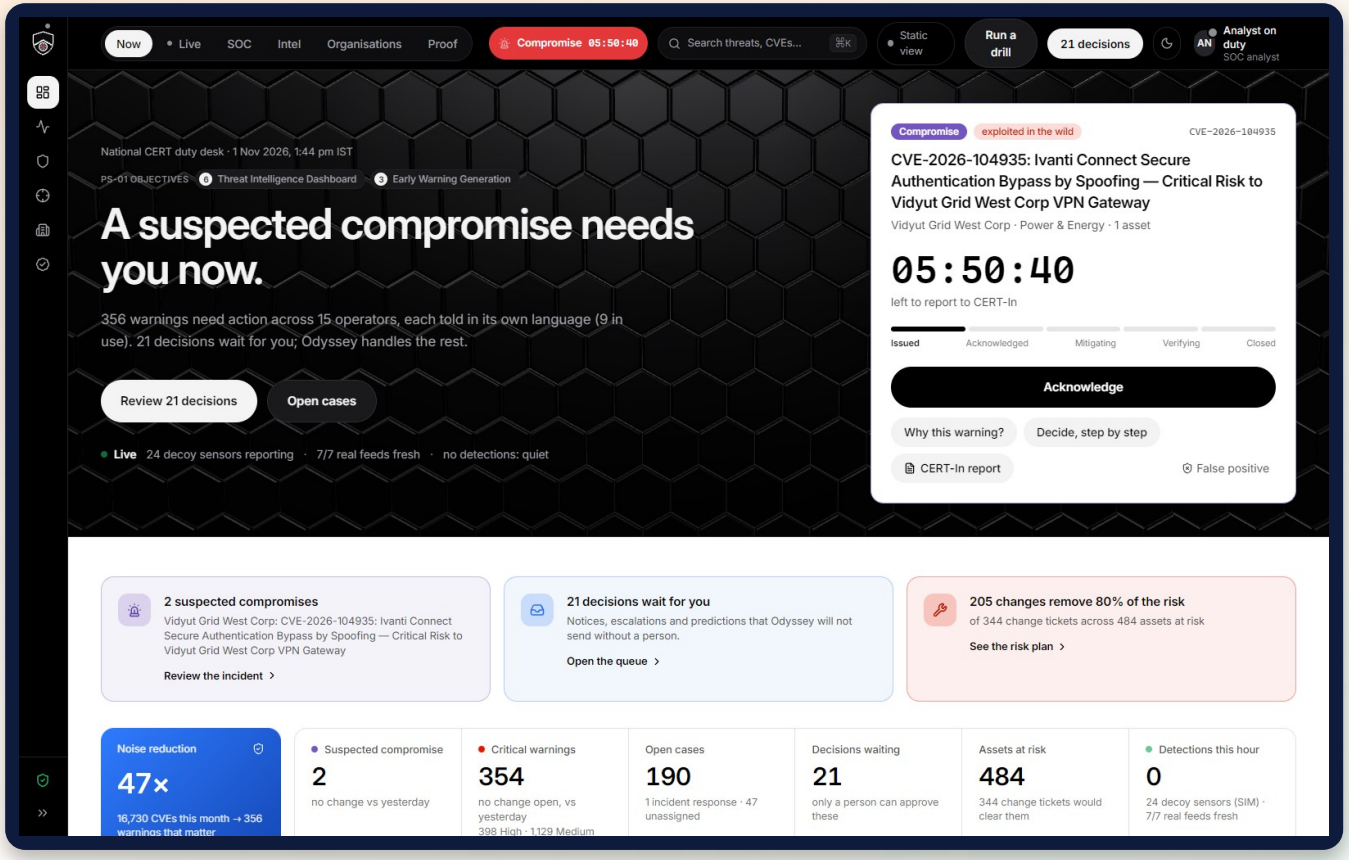


Intel › Advisories: the real CISA advisories matched to the inventory; the WinCC one touches 13 assets at 7 operators and proposes notices in their languages.

Evidence-based warnings go out on their own; predictions and escalations wait for a person. The time from publication to warning is a ledger event, shown on the Live page.

Legend PS-01: Problem Statement 01 · CVE: Common Vulnerabilities and Exposures · NVD: National Vulnerability Database (NIST) · CISA: Cybersecurity and Infrastructure Security Agency (US) · SSVC: Stakeholder-Specific Vulnerability Categorization · CERT-In: Indian Computer Emergency Response Team · ICS: Industrial Control Systems

The duty desk: what needs a person right now, with its clock and its evidence.



47×

less noise: 16,730 CVEs this month → 356 warnings

6 h

CERT-In clock on the suspected compromise

26

decisions waiting for a person

205

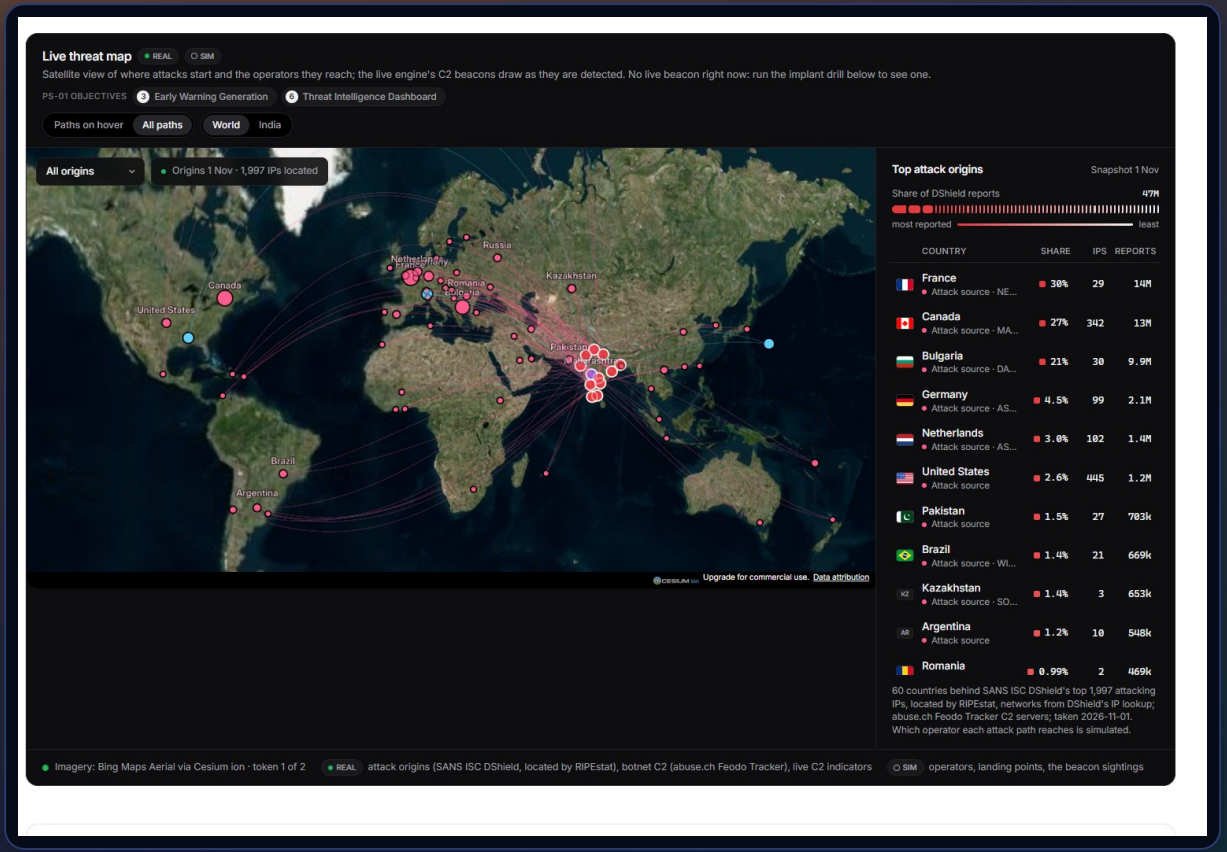
changes remove 80% of the risk

Now page: the suspected compromise with CERT-In's 6-hour clock, the decisions waiting, the risk plan and the noise reduction, every figure live from the engine.

Legend PS-01: Problem Statement 01 · CERT-In: Indian Computer Emergency Response Team

OBJECTIVES 3 AND 6 · EARLY WARNING ON A SATELLITE MAP

Where attacks start, whom they reach, and the live beacons as they are detected.



60

attacking countries, REAL (DShield, RIPEstat)

1,997

of the top 2,000 attacking IPs located

5

botnet C2 servers, Feodo Tracker

2 min

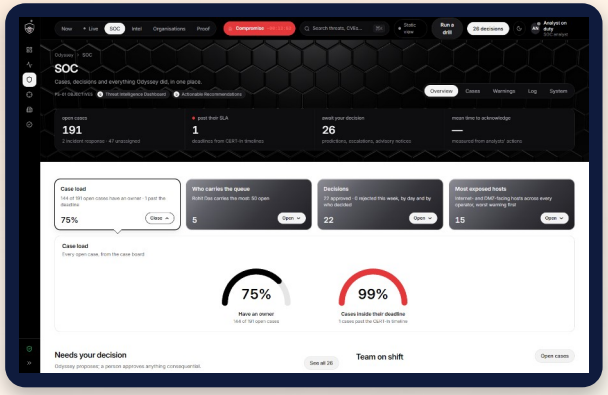
between NVD reads: every new CVE, ranked

Live page: a flat satellite world (CesiumJS; Sentinel-2 or Bing via Cesium ion, bundled tiles offline) with DShield's top attacking countries, Feodo C2s, the operators' states and attack paths as comet trails.

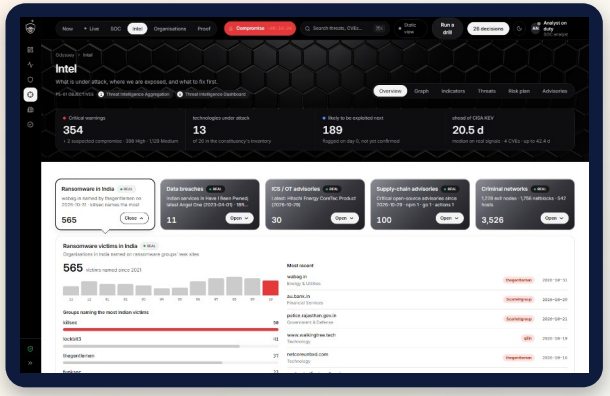
Legend PS-01: Problem Statement 01 · CVE: Common Vulnerabilities and Exposures · NVD: National Vulnerability Database (NIST) · C2: Command and Control (attacker infrastructure) · DShield: SANS Internet Storm Center's distributed sensor network · REAL: real public data

THE CONSOLE, END TO END

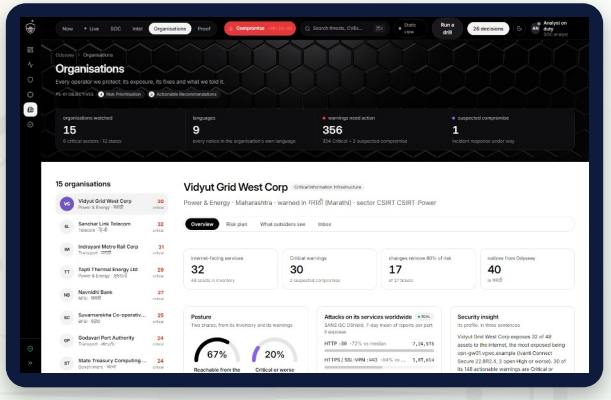
Six destinations, one question each; details slide in as sheets.



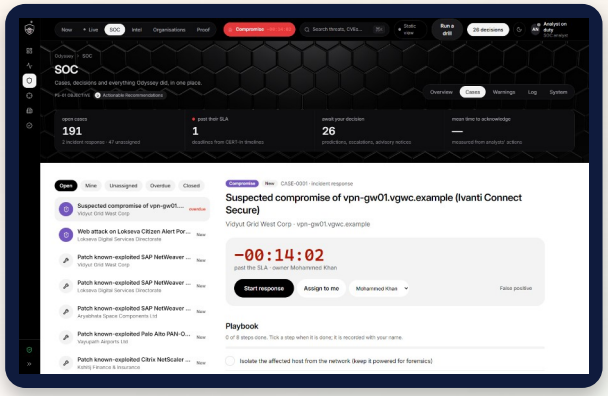
SOC › Overview: cases, SLAs, decisions



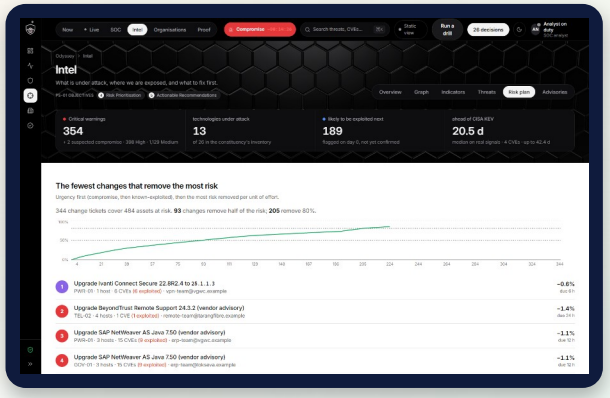
Intel › Overview: sources and real feeds



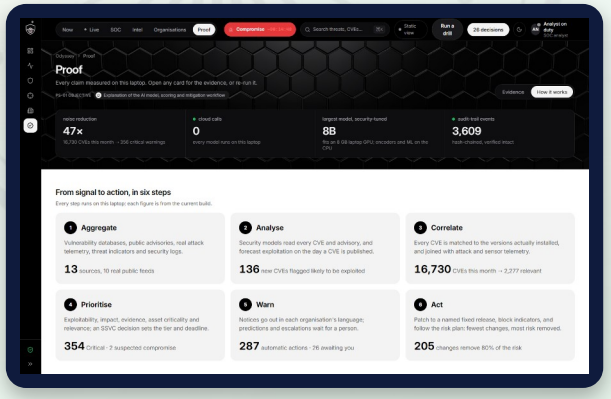
Organisations: one operator's picture



SOC › Cases: playbooks, MTTA, MTTR



Intel › Risk plan: fewest changes, most risk



Proof › How it works: models and scoring

Legend PS-01: Problem Statement 01 · SOC: Security Operations Centre · MTTA: Mean Time To Acknowledge · MTTR: Mean Time To Resolve

HOW IT WORKS

Four innovations, every one measured on this laptop.

- Risk prioritisation you can recompute
- Early warning before the incident
- Human in the loop, by construction
- Sovereign and offline



One transparent score, recomputable from the evidence, with India's clocks attached.

Risk = 100 · R · K · (0.55 L + 0.30 S + 0.15 V)

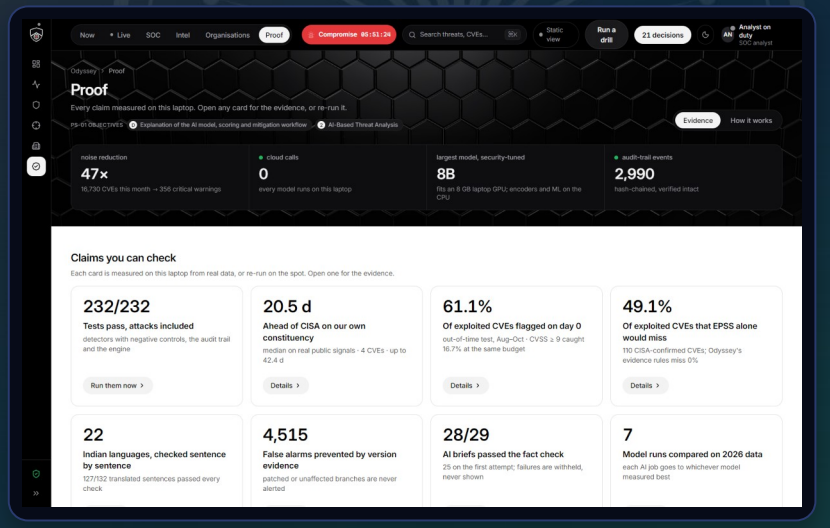
Term	Meaning	Where it comes from
L	Exploitability	Likelihood rules with floors: CISA KEV, SSVC active, decoy exploit hits, vendor "exploited" text, EPSS, the forecast
S	Potential impact	SSVC technical impact, OT safety minimum, CVSS
V	Evidence	Admiralty-graded sources, independent origins, decay with age
K	Asset context	Criticality tier, internet exposure, blast radius to crown jewels
R	Relevance	Installed version against NVD, the CNA record and advisory text; disputes go to an analyst

An SSVC decision sets the tier; a Bayesian P(compromise) orders a hunt at 20% and declares a suspected compromise at 50%, starting CERT-In's 6-hour clock. Every warning stores the hash of the scoring config, so any score can be recomputed.

47× less noise

measured this morning

16,730 CVEs published this month became 356 warnings that matter, 2 of them suspected compromises.



Proof › Evidence: every claim as a card, with its measurement.

Legend PS-01: Problem Statement 01 · NVD: National Vulnerability Database (NIST) · CISA: Cybersecurity and Infrastructure Security Agency (US) · KEV: Known Exploited Vulnerabilities catalogue · SSVC: Stakeholder-Specific Vulnerability Categorization · EPSS: Exploit Prediction Scoring System · CVSS: Common Vulnerability Scoring System · CNA: CVE Numbering Authority · CERT-In: Indian Computer Emergency Response Team · OT: Operational Technology

Four ways Odyssey gets ahead of the attacker, each one measured.

Live intake, every two minutes

Every CVE NVD publishes or changes is read, matched to installed versions, scored and ranked for the analyst with its reasons; an exposed match gets a notice within seconds.

Day-0 exploitation forecast

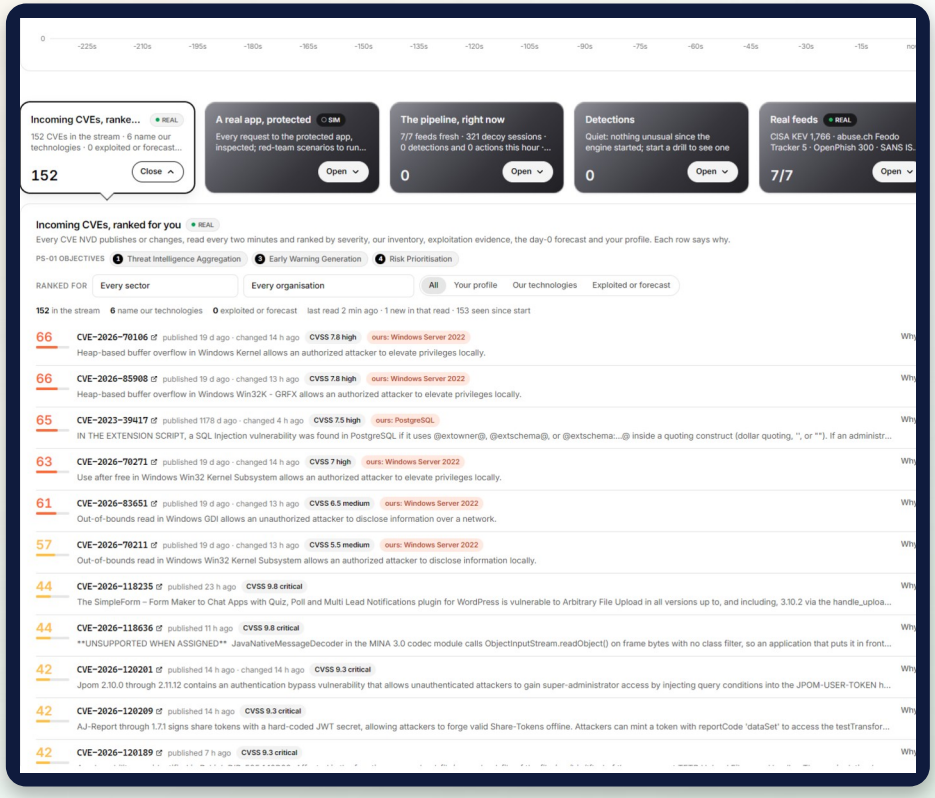
Logistic regression, trained 2021 to 2025 and frozen 31 July 2026. Out of sample it flags 2.0% of new CVEs and catches 61.1% of those later exploited; CVSS ≥ 9 catches 16.7%. ROC AUC 0.926.

Pre-disclosure scanning surges

A national decoy grid (24 sensors, SIM, calibrated to published honeypot patterns) raises a robust-z surge per product; the Ivanti surge came 12 days before disclosure.

Three live detectors, re-tuned by analysts

Surge, C2 beacon periodicity and an IsolationForest flow anomaly every five seconds; a beacon re-scores every warning on that host at once. Thresholds re-derive from the analysts' own decisions.



Live page: the incoming CVE stream, ranked for the analyst with the reasons for each rank, NVD read every two minutes.

Forecast precision is about 5% (exploitation is rare, base rate 0.16%), so forecasts are proposals for a person.

Legend PS-01: Problem Statement 01 · CVE: Common Vulnerabilities and Exposures · NVD: National Vulnerability Database (NIST) · CVSS: Common Vulnerability Scoring System · C2: Command and Control (attacker infrastructure) · ROC AUC: Receiver Operating Characteristic, Area Under the Curve · SIM: simulated data, labelled as such

INNOVATION 3 · HUMAN IN THE LOOP, BY CONSTRUCTION

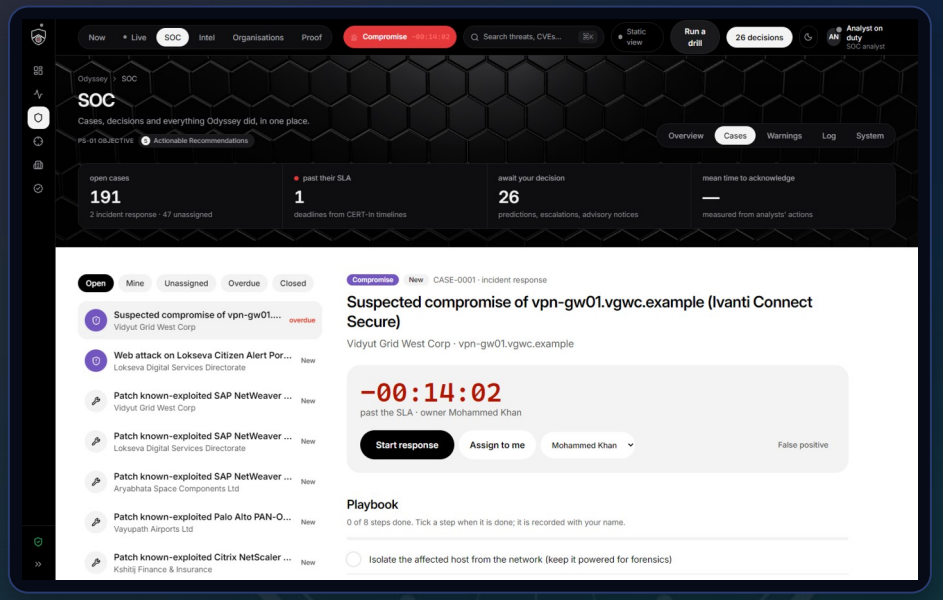
Odyssey acts on evidence, proposes on prediction, and never acts alone on anything consequential.

Runs on its own

- Notices on confirmed exploitation of an affected version
- Daily digests and threat hunts
- CERT-In report drafts, pre-filled

Waits for a person

- Day-0 predictions and sensor-only warnings
- Escalations to CERT-In, NCIIPC, sector CSIRTs
- Every rejection is recorded for calibration



SOC › Cases: the incident case opened by the detector, its playbook, owner and SLA; every step and note is a hash-chained ledger event.

A SOC on top: cases with owners and SLAs, playbooks, shift handover, MTTA and MTTR measured from what analysts actually did. A profile (sector, organisation) re-ranks the incoming stream and the recommendations for this analyst. Alerts reach Slack, Teams, Discord, e-mail or any webhook, one card per event, audited.

Legend PS-01: Problem Statement 01 · CERT-In: Indian Computer Emergency Response Team · NCIIPC: National Critical Information Infrastructure Protection Centre · SOC: Security Operations Centre · SLA: Service Level Agreement · MTTA: Mean Time To Acknowledge · MTTR: Mean Time To Resolve

From evidence to an audited decision in four steps, on every warning.

1

Evidence

Tier, risk, the rules that fired with their sources, exposure votes, the clocks that apply, live detections on the host.

2

Options

Patch, mitigate, hunt, escalate, verify, accept: each with its steps and effects, and Odyssey's recommendation with the reason.

3

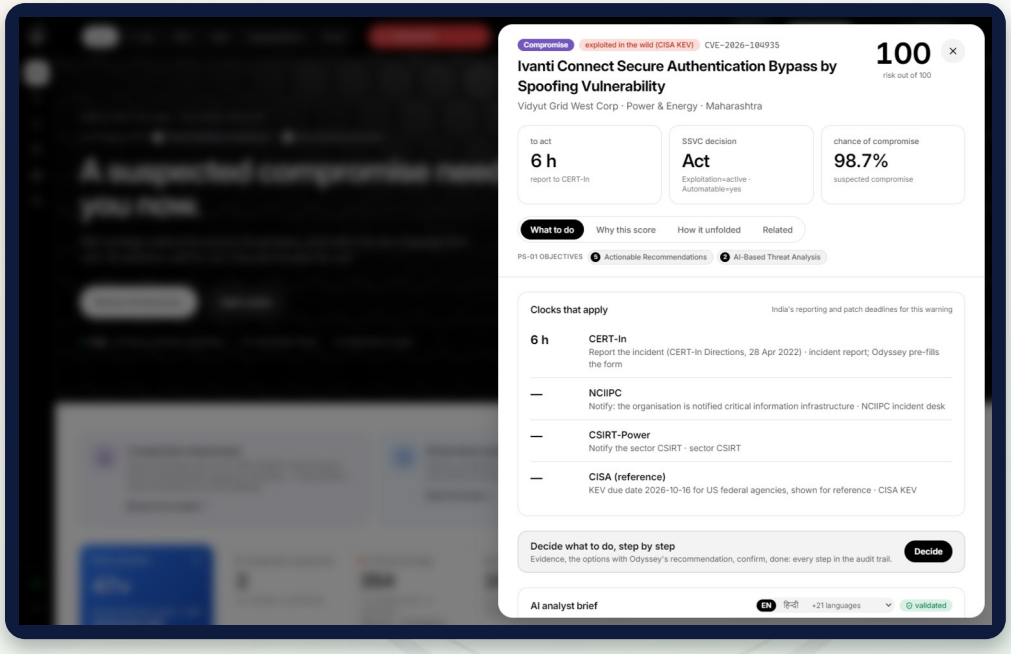
Confirm

Who to escalate to (CERT-In, the sector CSIRT, NCIIPC), a notice to the operator in its own language, a note for the audit trail.

4

Done

Workflow moved, case opened or advanced with its playbook, escalation card to the channels, every step returned with its ledger reference.

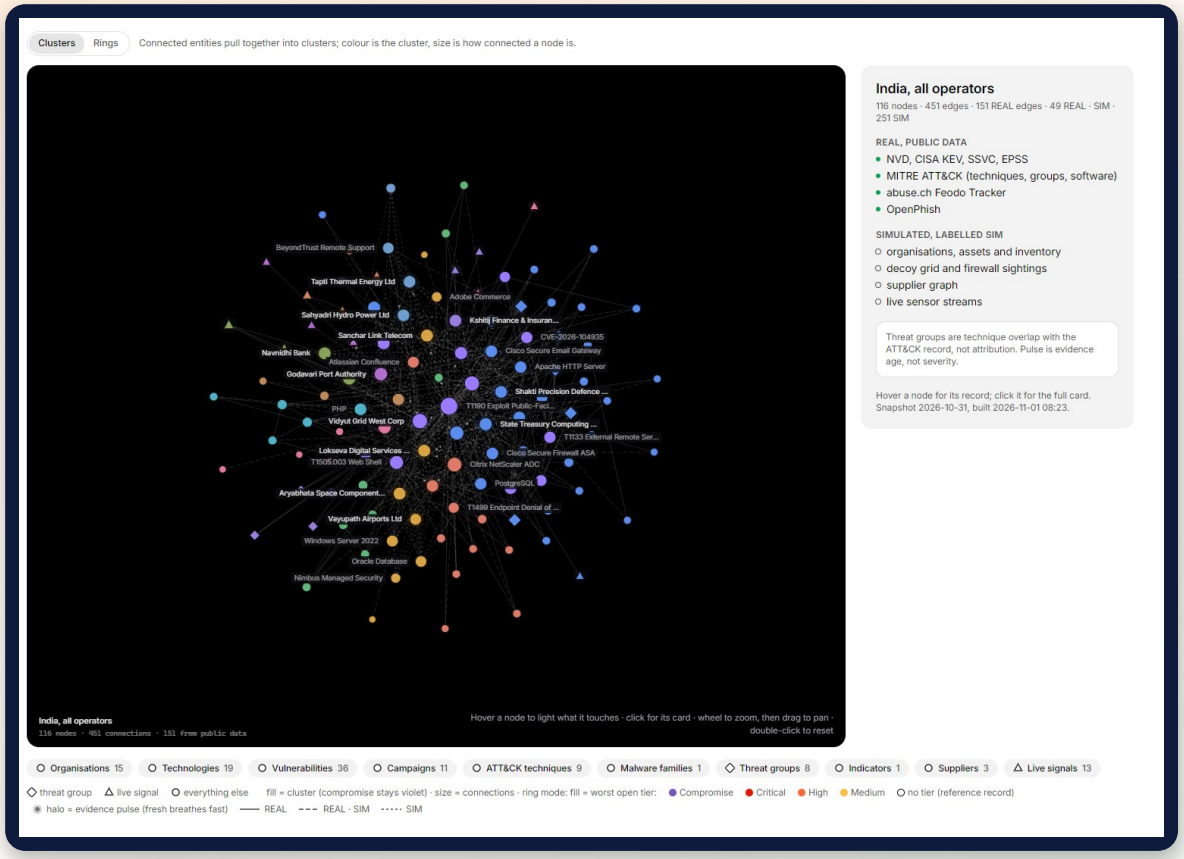


The warning sheet: tier, risk, clocks, the validated AI brief, the named fixed release, indicators to block, hunt steps, and "Decide, step by step".

Nothing runs until the analyst presses the last button, and then only through mechanisms that already existed and were tested: workflow, cases, channels, ledger.

Legend PS-01: Problem Statement 01 · CERT-In: Indian Computer Emergency Response Team · NCIIPC: National Critical Information Infrastructure Protection Centre · CSIRT: Computer Security Incident Response Team

The intelligence graph relates everything Odyssey knows, breathing with fresh evidence.



116

entities in the record, every edge with provenance

9

earlier instants the record can be replayed at

Top 5

threat groups by ATT&CK overlap, recommended with reasons

15 s

between watchlist evaluations

Intel › Graph: organisations, assets, technologies, CVEs, campaigns, ATT&CK techniques, malware, threat groups (overlap, never attribution), indicators and live signals; replay and watchlists.

Legend PS-01: Problem Statement 01 · ATT&CK: MITRE Adversarial Tactics, Techniques and Common Knowledge

INNOVATION 4 · SOVEREIGN AND OFFLINE

Everything runs on one laptop. Nothing leaves it unless an analyst routes it.

0

cloud calls

Every model, score and brief is computed here.

8 GB

of laptop GPU

One 8B security LLM resident; encoders and ML on the CPU.

13

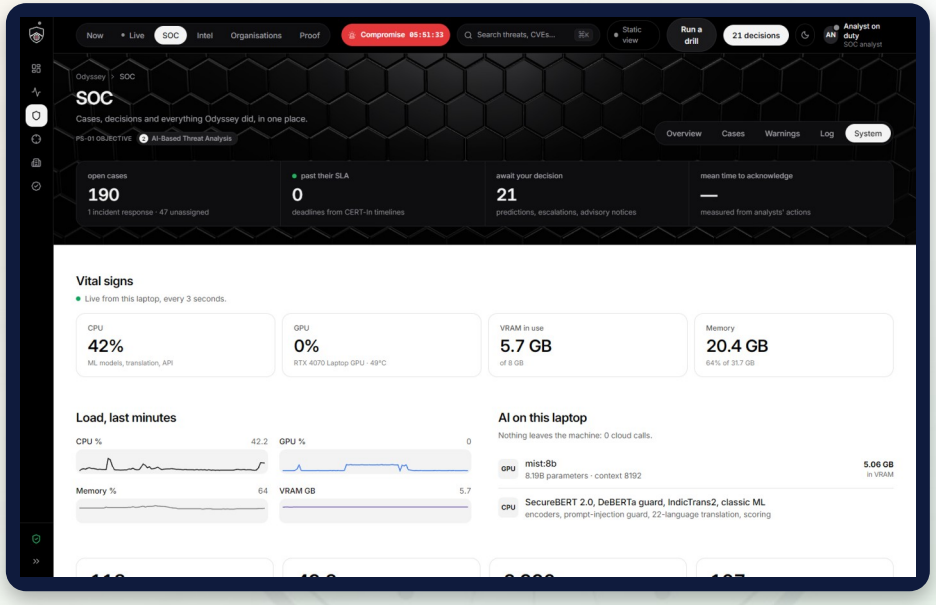
sources, snapshot-backed

Re-read live when online, written through to the snapshot.

22

scheduled languages

IndicTrans2 locally; 127 of 132 sentences validated.



SOC › System: the laptop's live vital signs, the models resident on the GPU, every AI job timed, the audit trail growing.

LIVE re-reads NVD every two minutes, KEV, Feodo, OpenPhish and SACHET every fifteen, DShield and CISA advisories hourly; DEMO serves the same screens from the snapshot. Every feed read, model output and click is a hash-chained ledger event.

Legend PS-01: Problem Statement 01 · NVD: National Vulnerability Database (NIST) · CISA: Cybersecurity and Infrastructure Security Agency (US) · KEV: Known Exploited Vulnerabilities catalogue · SACHET: NDMA's national alert dissemination platform · SOC: Security Operations Centre · LLM: Large Language Model · ML: Machine Learning · DShield: SANS Internet Storm Center's distributed sensor network

THE MODEL STACK · ALL LOCAL, ALL MEASURED

Security-tuned models explain; a deterministic engine decides.

Job	Model	Measured on 2026 data
Extract, classify, narrate	MiST 8B, security-tuned, Apache-2.0 (GPU)	79.9% vendor accuracy on every 2026 KEV CVE; best LLM on CWE
Deep reasoning on Critical alerts	Foundation-Sec 8B Reasoning, Cisco (GPU)	Security reasoning traces, in batches
Entities and related threats	SecureBERT 2.0 NER and bi-encoder (CPU)	Advisory entities; related-threat retrieval
Prompt-injection guard	DeBERTa-v3 prompt-injection v2 plus rules (CPU)	A hostile advisory is blocked on stage
22 Indian languages	IndicTrans2 en-indic 200M, AI4Bharat (CPU)	127 of 132 sentences validated
CWE and CVSS prediction	TF-IDF, LinearSVC and logistic heads (CPU)	CWE 77.5% (top-3 87.5%); CVSS metrics 80 to 98%
Day-0 exploitation forecast	Logistic regression, leak-free KEV history (CPU)	Flags 2.0%, catches 61.1%; ROC AUC 0.926
Live detection and decision	Robust-z, RITA-style beacon, IsolationForest, Bayes (CPU)	0 alerts in 240 quiet sensor-hours; drills caught



8B

the largest model, security-tuned, fits an 8 GB GPU

1 per role

chosen by local benchmarks, swappable by config

0

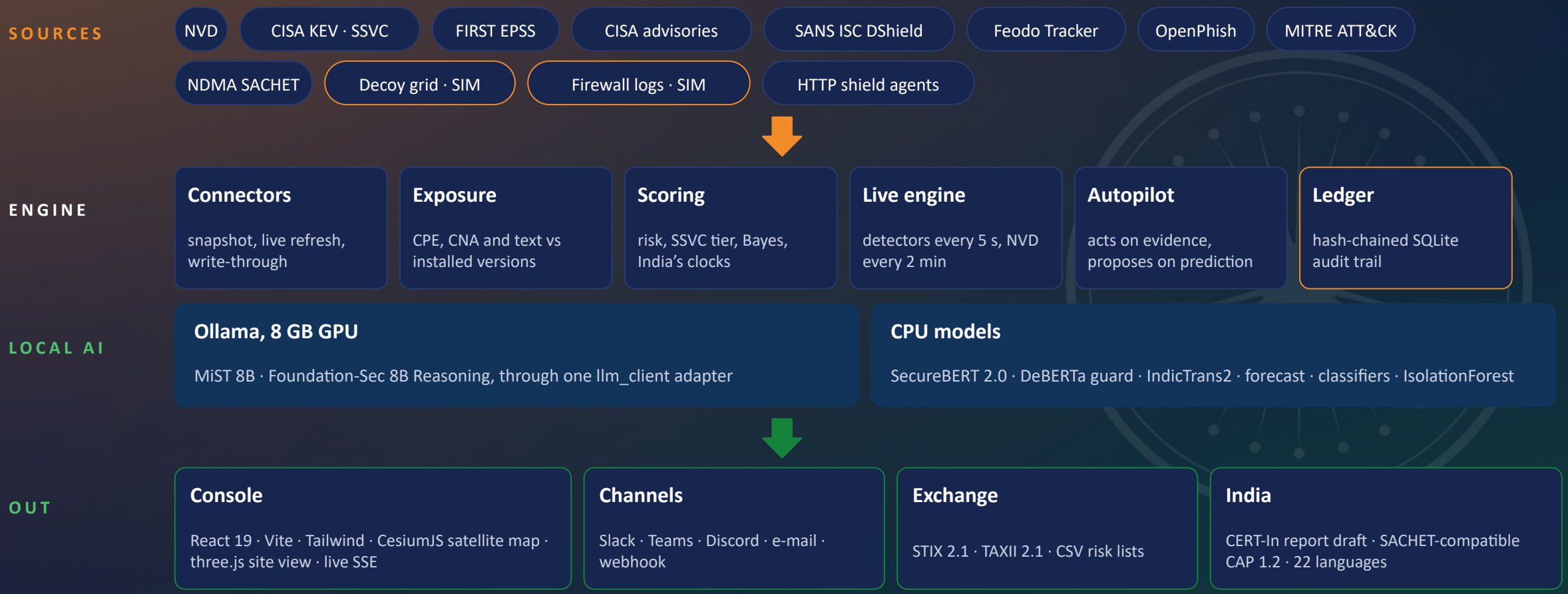
general LLMs in the running system: baselines only

Every brief is checked against the record before it is shown: a number the source never stated is caught, not published.

Legend PS-01: Problem Statement 01 · CVE: Common Vulnerabilities and Exposures · KEV: Known Exploited Vulnerabilities catalogue · CVSS: Common Vulnerability Scoring System · CWE: Common Weakness Enumeration · LLM: Large Language Model · NER: Named Entity Recognition · ROC AUC: Receiver Operating Characteristic, Area Under the Curve · RITA: Real Intelligence Threat Analytics (beacon analysis)

ARCHITECTURE AND TECH STACK

From thirteen sources to one audited decision, on one machine.



FastAPI 0.162 · Python 3.15 · SQLite · scikit-learn · PyTorch CPU · Ollama · pinned versions · one command starts the whole stack

Legend PS-01: Problem Statement 01 · NVD: National Vulnerability Database (NIST) · CISA: Cybersecurity and Infrastructure Security Agency (US) · KEV: Known Exploited Vulnerabilities catalogue · SSVC: Stakeholder-Specific Vulnerability Categorization · EPSS: Exploit Prediction Scoring System · FIRST: Forum of Incident Response and Security Teams · CPE: Common Platform Enumeration · CNA: CVE Numbering Authority · ATT&CK: MITRE Adversarial Tactics, Techniques and Common Knowledge · STIX: Structured Threat Information Expression · TAXII: Trusted Automated Exchange of Intelligence Information · CAP: Common Alerting Protocol · SACHET: NDMA's national alert dissemination platform · NDMA: National Disaster Management Authority · CERT-In: Indian Computer Emergency Response Team · DShield: SANS Internet Storm Center's distributed sensor network · SANS ISC: SANS Internet Storm Center · SIM: simulated data, labelled as such

EXPECTED DELIVERABLES

All four deliverables, each with its proof a click away.

A functional prototype

Offline on an 8 GB laptop GPU. About 140 engine tests, 26 product flows through the API, a 25-step browser audit, a test bench of 135 checks the judges can re-run.

A dashboard of threats, risk levels and alerts

Six destinations; every warning with its tier, risk, clocks and evidence; every page names the objective it serves.

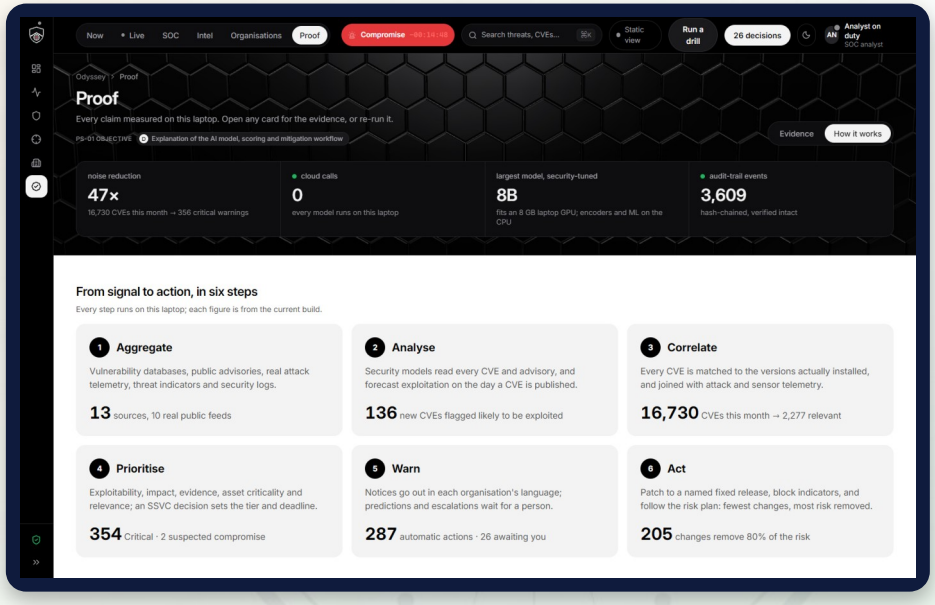
A demonstration on public intelligence and a controlled dataset

REAL: CVEs, KEV, SSVC, EPSS, advisories, DShield, Feodo, phishing, SACHET. SIM: 15 operators, the decoy grid, firewall logs. Every item labelled.

An explanation of the AI, the scoring and the mitigation workflow

Proof › How it works walks the pipeline, the model roster, the formula on a live example and the workflow; model card and coverage documents in the repository.

Red-teamed: 45 real web attacks at the lab app, 45 detected and blocked, 0 false positives; 10 hostile CAP messages, 10 refused, no leak.



Proof › How it works: from signal to act in six steps, the models with measured results, the scoring on a live example.

Legend PS-01: Problem Statement 01 · KEV: Known Exploited Vulnerabilities catalogue · SSVC: Stakeholder-Specific Vulnerability Categorization · EPSS: Exploit Prediction Scoring System · CAP: Common Alerting Protocol · SACHET: NDMA's national alert dissemination platform · DShield: SANS Internet Storm Center's distributed sensor network · SIM: simulated data, labelled as such · REAL: real public data

WHAT WE SAY BEFORE YOU ASK

Honest limits, stated on the screens themselves.

Sensors are simulated

The decoy grid and firewall flows are SIM, labelled, accelerated; the feeds beside them are real and live.

Forecast precision is low

About 5%, because exploitation is rare (base rate 0.16%); forecasts are proposals for a person.

Lead time has two origins

Real signals and the simulated decoy grid; the Proof page attributes each lead to one of them.

Groups are overlap, not attribution

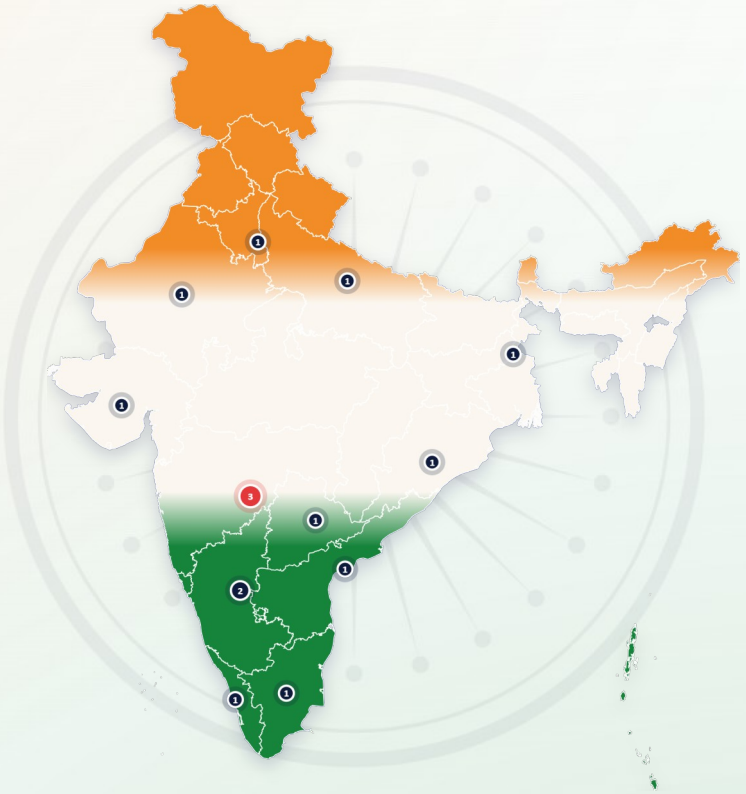
A threat group appears only as overlap with its public ATT&CK record. Odyssey never attributes activity.

Translation is checked, not assumed

127 of 132 sentences validated across 22 languages; Santali is weakest; withheld lines fall back to English.

CERT-In has no machine-readable feed

Its advisories are style-matched samples, labelled SIM; the real national feed we read is NDMA's SACHET.



Every fact on screen is labelled REAL or SIM; every claim on the Proof page is measured on this laptop and can be re-run.

Legend PS-01: Problem Statement 01 · ATT&CK: MITRE Adversarial Tactics, Techniques and Common Knowledge · SACHET: NDMA's national alert dissemination platform · NDMA: National Disaster Management Authority · CERT-In: Indian Computer Emergency Response Team · SIM: simulated data, labelled as such · REAL: real public data

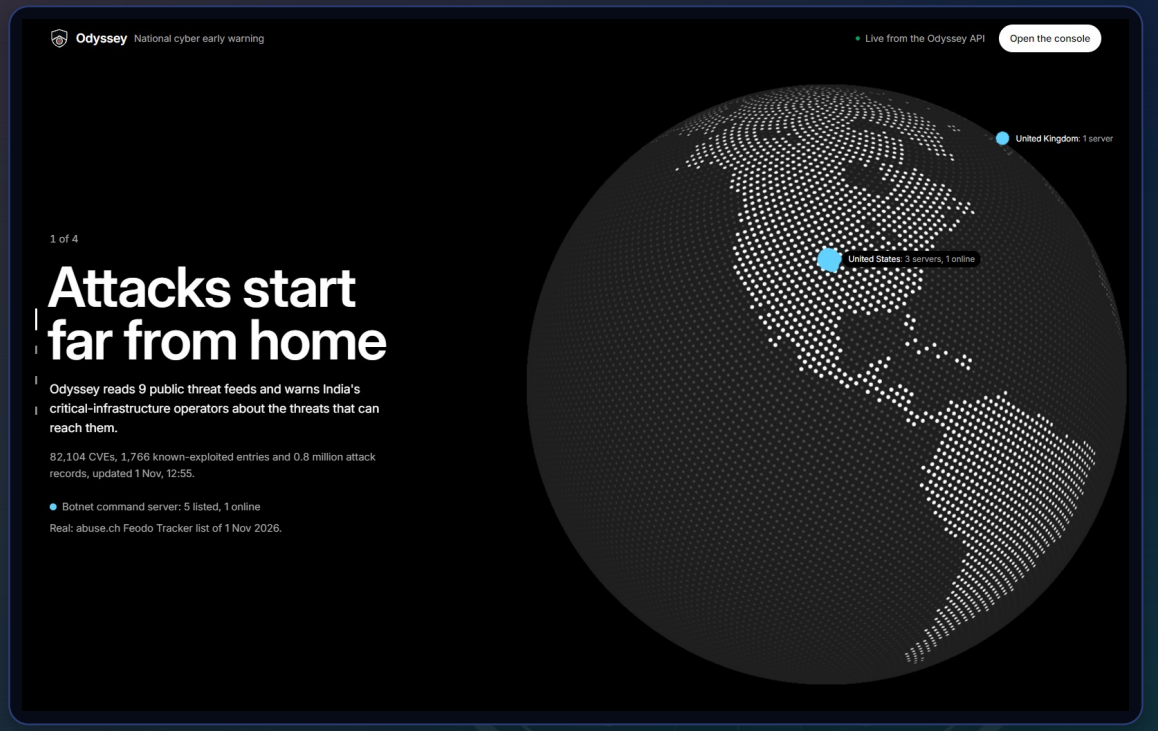
JOIN THE HUNT

Early warning for the nation, from one laptop.

Live demo on this machine: the duty desk, the satellite threat map, a real CVE taken in from NVD, a red-team against the shield, and one decision taken step by step.

Aggregate · Analyse · Warn · Prioritise · Act · See

Team Odyssey · D3COY 1.0 · PS-01 · Thank you



Legend PS-01: Problem Statement 01 · CVE: Common Vulnerabilities and Exposures · NVD: National Vulnerability Database (NIST)